FIITJEE

ALL INDIA TEST SERIES

OPEN TEST

JEE (Main)-2021

TEST DATE: 10-01-2021

ANSWERS, HINTS & SOLUTIONS

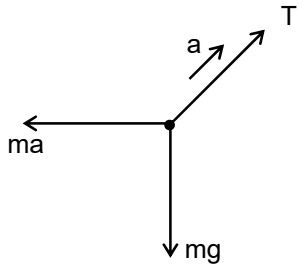
Physics

PART – I

SECTION – A

1. C

Sol. $T - m\sqrt{g^2 + a^2} = ma$



2. D

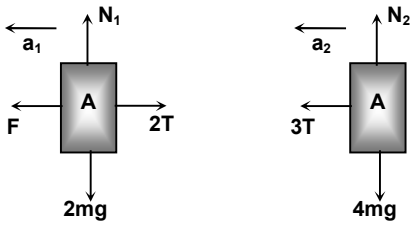
Sol. $F - 2T = 2ma_1$

$3T = 4ma_2$

$2a_1 = 3a_2$

Solving we get

$a_2 = \frac{3F}{17m}$



3. B

Sol. $(m_1 + m_2)g\left(1 - \frac{\rho}{\sigma}\right) - 6\pi\eta(r_1 + r_2)V = 0$

$m_1 g - \frac{m_1\rho}{\sigma}g - T - 6\pi\eta r_1 V = 0$

$$T = m_1 g\left(\frac{\sigma-\rho}{\sigma}\right) - \frac{6\pi\eta r_1(m_1+m_2)g\left(\frac{\sigma-\rho}{\sigma}\right)}{6\pi\eta(r_1+r_2)}$$

$$T = \frac{4}{3}\pi r_1 r_2 (r_1 - r_2) g(\sigma - \rho)$$

4. B

Sol. $y = \frac{\Delta\ell_1 + \Delta\ell_2}{2} = \frac{T}{2AY}(\ell_1 + \ell_2)$

$= \frac{5\times10^3(600+900)\times10^{-3}}{2\times10^{-3}\times10^9} = 3.75 \text{ mm}$

5. C

Sol. $15\sigma Vg - \sigma Vg - w = \left(\frac{w}{g} + \sigma V\right)a$

$$a = \left(\frac{14\sigma Vg - w}{w + \sigma Vg}\right)g$$

6. D

Sol. T = ma

$$T = 2\times\frac{F}{5}$$

$$0.25\times10^9\times0.4\times10^{-6} = \frac{2F}{5}$$

$$10^2 = \frac{2F}{5}$$

$$F_{max} = \frac{5}{2}\times10^2 = 250 \text{ N}$$

7. A

Sol. $Q = Q_0 \sin(\omega t + \phi)$

$$i = \frac{dQ}{dt} = Q_0\omega\cos(\omega t + \phi)$$

At t = 0

$4 = Q_0 \sin\phi$

$$-\sqrt{5} = \frac{Q_0}{2}\cos\phi$$

$16 + 20 = Q_0^2$

Maximum charge, $Q_0 = 6C$

$$\phi = \pi - \sin^{-1}\left(\frac{2}{3}\right)$$

8. C

Sol. $$C = \frac{C_1C_2}{C_1+C_2} = \frac{\frac{A\varepsilon_0}{x}\times\frac{A\varepsilon_0}{(a-b-x)}}{\frac{A\varepsilon_0}{x}+\frac{A\varepsilon_0}{(a-b-x)}}$$

$$C = \frac{A\varepsilon_0}{(a-b)} \text{ independent of x}$$

9. A

Sol. $\Delta W_{cycle} = \frac{1}{2}V_0P_0 = \frac{P_0V_0}{2}$

For the process AB, P= KV $\Rightarrow$ PV^{-1} = constant

Molar heat capacity of the gas in the process AB,

$C = C_V + \frac{R}{(1-x)} = \frac{3R}{2} + \frac{R}{2} = 2R$

$\Delta Q_{AB} = nC\Delta T = n2R(4T_0 - T_0) = 6nRT_0 = 6P_0V_0$

$\Delta Q_{BC} < 0$ and $\Delta Q_{CA} < 0$

The efficiency of the cyclic process, $\eta = \frac{\Delta W_{cycle}}{\Delta Q_{supplied}} \times 100 = \frac{P_0V_0}{2 \times 6P_0V_0} \times 100$

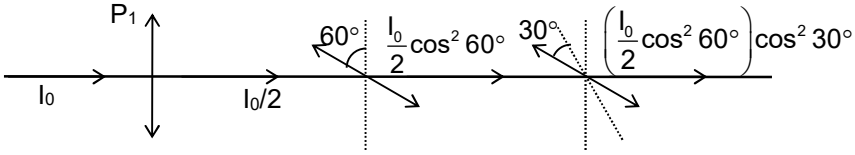
$\eta = \frac{25}{3} = 8.33\%$

10. A

Sol. By the symmetry, $\vec{B}_{total} = 0$

11. B

Sol. $I' = \frac{I_0}{2} \times \frac{1}{4} \times \frac{3}{4} = \frac{3I_0}{32}$



12. B

Sol. λ_{min} = 66.3 pm

$V = \frac{1240}{66.3 \times 10^{-3}} = 18.70$ kV

de-Broglie wavelength for electron = $\frac{h}{\sqrt{2meV}} = 0.09$ Å

13. B

Sol. $\mu = \frac{e}{T}\pi r^2 = \frac{e\omega\pi r^2}{2\pi} = \frac{er^2\omega}{2}$

$L = mr^2\omega$

$\frac{\mu}{L} = \frac{e}{2m}$

14. C

Sol. At 4t its $\left(\frac{1}{2}\right)^4 N_0$

So, at 5t it will be $\left(\frac{1}{2}\right)^5 N_0$

i.e., $t = T_{1/2}$

15. C

Sol. $y = \frac{x^2}{4a}$

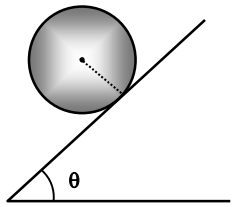
$\frac{dy}{dx} = \frac{2x}{4a}$

When y = a, $\frac{dy}{dx} = 1 = \tan\theta$

Tangential acceleration = g sin θ = $\frac{g}{\sqrt{2}}$

16. D

Sol. $\because \mu = \tan\theta$



Till the cylinder rotates it will remain stationary.

as $mg\sin\theta = \mu mg\cos\theta$

$\mu mg\cos\theta R = \frac{mR^2}{2}\alpha$

$\alpha = \frac{mg\sin\theta R \cdot 2}{mR^2} = \frac{2g\sin\theta}{R}$

$\omega_0 = \alpha t \Rightarrow t = \frac{\omega_0}{\alpha} = \frac{\omega_0 R}{2g\sin\theta}$

17. C

Sol. Q = (80 × 7) + (120 × 8) – (200 × 6.5) MeV = 220 MeV

18. A

Sol. Velocity of longitudinal wave in the rod,

$v = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{2\times10^{11}}{8\times10^{3}}} = 5000\ \text{m/s}$

Now,

$n_1 : n_2 : n_3 = \ell_1 : \ell_2 : \ell_3$

= 25 : 40 : 35 = 2.5 : 4 : 3.5

The minimum no of loops formed

n = 2.5 + 4 + 3.5 = 10 loops

$\frac{10\lambda}{2} = \ell \Rightarrow \lambda = 0.20\ \text{m}$

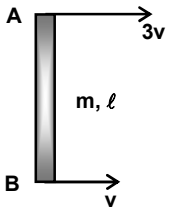
The minimum frequency of vibration, $f = \frac{v}{\lambda} = \frac{5000}{0.2} = 25\ \text{kHz}$

19. D

Sol. By the property of full wave rectifier

20. A

Sol. The angular velocity of the rod, $\omega = \frac{3v - v}{\ell} = \frac{2v}{\ell}$



The velocity of centre of mass of the rod,

$v_{CM} = \frac{\omega\ell}{2} + v = 2v$

The kinetic energy of the rod, $K = \frac{1}{2}\frac{m\ell^2}{12}\omega^2 + \frac{1}{2}mv_{CM}^2$

$K = \frac{mv^2}{6} + \frac{1}{2}m(2v)^2 = \frac{13}{6}mv^2$

SECTION – B

21. 2

Sol. $-\frac{dN_A}{dt} = \lambda N_0 e^{-\lambda t}$

$-\frac{dN_B}{dt} = 2\lambda N_0' e^{-2\lambda t}$

At t = 0, $4\frac{dN_A}{dt} = \frac{dN_B}{dt}$ (20% and 80%)

$N_0 \lambda e^{-\lambda t} = 2N_0 \cdot 2\lambda e^{-2\lambda t}$

$t = \frac{2\ell n2}{\lambda} = 2$ hours

22. 5

Sol. $iR = 4V$

$L\frac{di}{dt} = 8V$

$L\frac{di}{dt} i = \frac{16}{x}$

$x = \frac{16}{3.2} = 5$

SECTION – C

23. 00037.50

Sol. $\frac{1}{f_1} = \left(\frac{3}{2} - 1\right)\left(\frac{1}{\infty} + \frac{1}{R}\right) = \frac{1}{2R}$

$2R = f_1 \Rightarrow R = \frac{f_1}{2} = \frac{25}{4}$ cm

$\frac{1}{f_2} = \left(\frac{4}{3} - 1\right)\left(\frac{1}{-R} - \frac{1}{\infty}\right) = -\frac{1}{3R}$

$f_2 = -3R = -\frac{75}{4}$ cm

Now, $-\frac{1}{F} = \frac{2}{f_1} + \frac{2}{f_2} - \frac{1}{f_m}$

$-\frac{1}{F} = \frac{4}{25} - \frac{8}{75} - \frac{1}{\infty}$

$-\frac{1}{F} = \frac{4}{75}$

$\Rightarrow F = -\frac{75}{4}$ cm

The point object is placed at a distance,

$d = \frac{75}{2} = 37.50$ cm

24. 00029.20

Sol. $\frac{2T}{r} = \rho g h$

$$h = \frac{73\times10^{-3}\times2}{\frac{0.1}{2}\times10\times10^{3}\times10^{-3}} = 29.2 \text{ cm}$$

25. 00006.72

Sol. $\frac{F}{A} = \frac{Y\Delta\ell}{\ell}$

$mg = YA\alpha\Delta t$

$mg = 1.2\times10^{11}\times2\times10^{-6}\times1.4\times10^{-6}\times200$

$mg = 67.2 \Rightarrow m = 6.72$ kg

Chemistry

PART – II

SECTION – A

26. B

Sol. I. $n + \ell = 4 + 1 = 5$

II. $n + \ell = 4 + 0 = 4$

III. $n + \ell = 3 + 2 = 5$

IV. $n + \ell = 3 + 1 = 4$

I is having greater n value as compared to III.
II is having greater n value as compared to IV.

27. C

Sol.

$$M \xrightarrow[HNO_3]{\text{dilute}} Zn(NO_3)_2 \xrightarrow{\text{aq. NaOH}} Zn(OH)_2$$

White precipitate

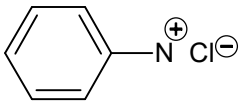
$\downarrow$ Excess NaOH

$$ZnS \downarrow \xleftarrow{H_2S} Na_2\left[Zn(OH)_4\right]$$

White precipitate

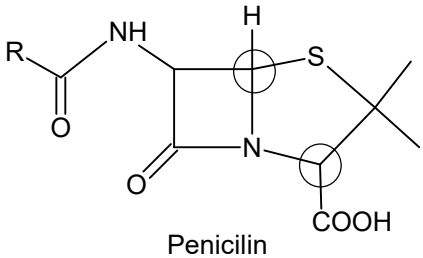
28. C

Sol. As it gives N_2 gas on heating, which escapes from reaction medium.



29. A

Sol.



Penicilin

30. A

Sol. $Ba(OH)_2(s) \rightleftharpoons Ba^{2+}(aq) + 2OH^-(aq)$

$pH = 12 \qquad \therefore pOH = 2$

i.e. $\left[OH^-\right] = 10^{-2}$

$$\left[Ba^{2+}\right] = \frac{10^{-2}}{2} = 5 \times 10^{-3}$$

$$K_{sp} = \left[Ba^{2+}\right]\left[OH^-\right]^2 = 5 \times 10^{-3} \times \left(10^{-2}\right)^2$$

$$= 5 \times 10^{-7}$$

31. C

Sol. Work done $= -P\Delta V = -1(2.5-2) = -1\times 0.5$

$= -0.5$ litre atm

$= -50.63$ J

$Q = \Delta U - W$

$300 = \Delta U - (-50.63)$

$\Delta U = 249.37$ J

32. C

Sol. The compound must be cyclic and contains a – OH and – CHO group. It must contain six C atoms.

33. B

Sol. $\underbrace{BrO_3^- \xrightarrow[n_1=5]{E_1^o} BrO^- \xrightarrow[n_2=2]{E_2^o} \frac{1}{2}Br_2}_{E^o}$

$$E^o = \frac{n_1E_1^o + n_2E_2^o}{n_1+n_2} = \frac{5\times 0.54 + 2\times 0.41}{5+2}$$

= 0.52 V

34. A

Sol. Amount of oxalic acid initially present $= \frac{1\times 50}{1000}\times 63 = 3.15$ gm

Amount of oxalic acid after adsorption $= \frac{50}{1000}\times\frac{63}{2} = 1.575$ gm

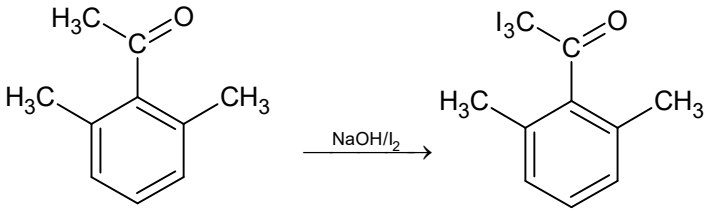
Amount adsorbed = 3.15 – 1.575 = 1.575 gm

Amount adsorbed per gram of charcoal = 1.575 × 2 = 3.15 gm.

35. B

Sol. $2Cu^{2+} + K_4\left[Fe(CN)_6\right] \longrightarrow 4K^+ + Cu_2\left[Fe(CN)_6\right]$

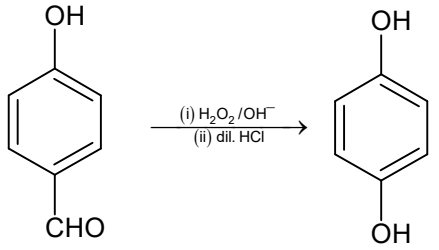
36. C

Sol. 

Nucleophilic attack by OH^- in the next step is not possible due to steric hindrance.

37. C

Sol.



P does not react with aq. $NaHCO_3$ as it is a weak acid.

38. B

Sol.

$$Ca(NO_3)_2 \rightleftharpoons Ca^{2+} + 2NO_3^-$$

Before dissociation	1	0	0
After dissociation	$1-\alpha$	α	2α

Total mole of at equilibrium $= 1-\alpha+\alpha+2\alpha = 1+2\alpha = 1+2\times 0.7 = 2.4$

$[\alpha = 0.7]$

$$\frac{m_{calc.}}{m_{exp}} = 1+2\alpha$$

$$m_{exp} = \frac{m_N}{1+2\alpha} = \frac{164}{2.4}$$

Also at $100^\circ C$, P of H_2O $P^o_{H_2O} = 760$ mm

$W_{solute} = 7$ gm $\quad W_{H_2O} = 100$ gm

$$\frac{P^o - P_s}{P_s} = \frac{7\times 18}{68.33\times 100} = 0.0184$$

$$P_s = \frac{760}{1.0184} = 746.27 \text{ mm}$$

39. B

Sol. In alkaline medium, chromate ions are present while in acidic medium, dichromate ions are present.

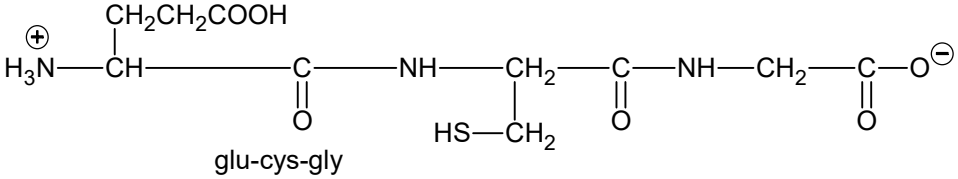
$$2CrO_4^{2-} \xrightarrow[PH=3]{H^+} Cr_2O_7^{2-} \xrightarrow[PH=8]{OH^-} 2CrO_4^{2-}$$

40. B

Sol. In this reaction HNO_3 behaves as H^+ acceptor and H_2SO_4 behaves as H^+ donor.

41. C

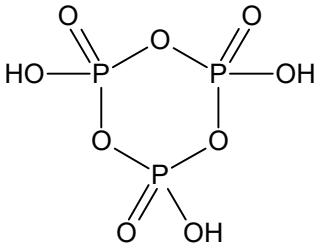
Sol. Since the tripeptide on hydrolysis gave two dipeptides glu-cys and cys-gly. Hence, cystine must be in between glutamic acid and glycine as given below



glu-cys-gly

42. C

Sol.



43. A

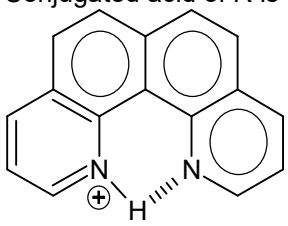
Sol. Due to +R effect of $-OCH_3$ group, it reacts with electrophile dichlorocarbene in fastest rate.

44. C

Sol. In $\left[V(CO)_6\right]^-$, the tendency of back donation of electron in π^* orbital of CO is greater, thus, V – C bond length decreases.

45. B

Sol. Conjugated acid of R is stabilized through intramolecular H bonding.



SECTION – B

46. 6

Sol. $\overset{(-2)}{Pb}\overset{}{S} \xrightarrow{\text{Roasting}} PbO + \overset{+4}{S}O_2$

Change of oxidation state = (4 – (–2)) = 6

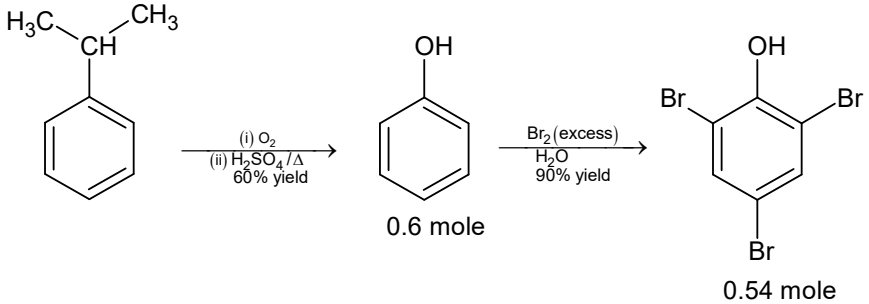
47. 6

Sol. Factual.

SECTION – C

48. 00172.26

Sol.



Mass of 5.4 mole 2,4, 6-Tribromo phenol = 172.26

49. 00028.95

Sol. $\Delta S = nF\left(\frac{dE}{dT}\right)_P = 2 \times 96500 \times 1.5 \times 10^{-4} = 28.95 \text{ J/K} - \text{mol}$

50. 00000.02

Sol. If P = initial partial pressure of NO_2

$$2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g)$$

$t = 0$	P		
t_{eqbm}	$P - 2x$	$2x$	x

$x = 0.25$ $\quad\therefore 2x = 0.5$

$$K_P = \frac{(2x)^2(x)}{(P-2x)^2} \Rightarrow 156.25 = \frac{(0.5)^2(0.25)}{P_{NO_2}^2}$$

$\therefore P_{NO_2} = 0.02$ atm

Mathematics

PART – III

SECTION – A

51. A

Sol. $T_r = \tan^{-1}\left(\frac{3}{9r^2-3r-1}\right) = \tan^{-1}\left(\frac{(3r+1)-(3r-2)}{1+(3r+1)(3r-2)}\right)$

$\sum_{r=1}^{n} T_r = \tan^{-1}(3n+1) - \tan^{-1}(1)$

$\sum_{r=1}^{\infty} T_r = \tan^{-1}(1)$

52. A

Sol. $S = \left(\frac{1-\frac{1}{2}}{1-\frac{1}{2}}\right) + \left(\frac{1-\frac{1}{2^2}}{1-\frac{1}{2}}\right)\left(\frac{1}{3}\right) + \left(\frac{1-\frac{1}{2^3}}{1-\frac{1}{2}}\right)\left(\frac{1}{3^2}\right) + ...$

$= 2\left\{\left(1+\frac{1}{3}+\frac{1}{3^2}+.....\right) - \frac{1}{2}\left(1+\frac{1}{2\cdot 3}+\frac{1}{(2\cdot 3)^2}+.....\right)\right\}$

$= 2\left\{\frac{1}{\left(1-\frac{1}{3}\right)} - \frac{1}{2}\cdot\frac{1}{\left(1-\frac{1}{6}\right)}\right\} = 2\left\{\frac{3}{2} - \frac{1}{2}\times\frac{6}{5}\right\} = 3 - \frac{6}{5} = \frac{9}{5}$

53. C

Sol. Both (a, b) and (b, a) lie on circle

54. D

Sol. $f(x) + f(1 - x) = 1$

55. C

Sol. Let number of red, white, blue and green balls are a, b, c and d respectively

$\therefore a + b + c + d = 18$

$a, b, c, d \geq 2$

Required number of way = $^{13}C_3 = 286$

56. C

Sol. Let, $x + y + z = \alpha$ and $\lambda = 5050$

$\cos x + \cos y + \cos z = \lambda \cos \alpha$

$\sin x + \sin y + \sin z = \lambda \sin \alpha$

$\cos(x + y) + \cos(y + z) + \cos(z + x) = \cos(\alpha - z) + \cos(\alpha - x) + \cos(\alpha - y)$

$= \cos \alpha(\lambda \cos \alpha) + \sin \alpha(\lambda \sin \alpha) = \lambda = 5050$

57. D

Sol. $\lim_{x\to 1} \frac{f'(1+x^3-x)(3x^2-1) - f'(x)}{1} = \lim_{x\to 0} \frac{-f'(1-x) - 0}{1} + 10$

$\Rightarrow f'(1) = 5$

58. D

Sol. Equation of line is $\frac{x-6}{\cos\theta}=\frac{y-8}{\sin\theta}=r$

$\therefore 2(r\cos\theta+6)^2+(r\sin\theta+8)^2=2$

$\Rightarrow (2\cos^2\theta+\sin^2\theta)r^2+2(12\cos\theta+8\sin\theta)r+134=0$

AB and AC are roots

AB, AP, AC are in H.P.

$\frac{2}{AP}=\frac{AB+AC}{AB\cdot AC}\Rightarrow\frac{1}{r}=\frac{-(6\cos\theta+4\sin\theta)}{67}$

$\Rightarrow -6r\cos\theta-4r\sin\theta=67\Rightarrow -6(x-6)-4(y-8)-67=0$

$\Rightarrow 6(x-6)+4(y-8)+67=0\Rightarrow 6x+4y-1=0$

Required distance = $\frac{1}{\sqrt{52}}$

59. D

Sol. Let $\lambda=\frac{225}{h^2}+\frac{25}{k^2}$

$h=5\cos\theta,\ k=5\sin\theta$

$\lambda=9\sec^2\theta+\text{cosec}^2\theta$

$\lambda_{min}=16$

60. B

Sol. $(a-1)^2+(b-1)^2+(c-2)^2=0$

$a=b=1,\ c=2$

$x^2+x+2=0$

(D < 0, therefore both the roots are common)

$\therefore\ \frac{2p}{1}=\frac{q}{1}=\frac{4r}{2}=\lambda(\text{say})$

$p=\frac{\lambda}{2},\ q=\lambda$

$r=\frac{\lambda}{2}$

$2p+q+3r=\frac{7}{2}\lambda$

$(2p+q+3r)_{min}=\frac{7}{2}\times 2=7$

61. D

Sol. Clearly C = 90°, A = 30°, B = 60°

$\frac{b-a}{b+a}=\frac{\sin B-\sin A}{\sin B+\sin A}=2-\sqrt{3}$

$\left(\left(\frac{b-a}{b+a}\right)x+\sqrt{3}y\right)^{10}=\left((2-\sqrt{3})x+\sqrt{3}y\right)^{10}$

$x=y=1$

$\left(2-\sqrt{3}+\sqrt{3}\right)^{10}=2^{10}$

62. B

Sol. $(x-x_0)(x^2-qx+2)=x^3+x^2-px+q$

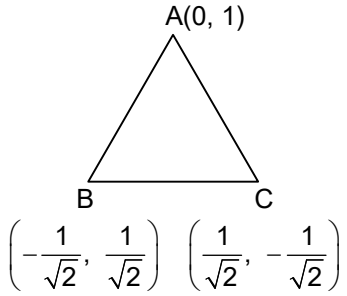
$-x_0-q=1$

$2+qx_0=-p$

$-2x_0 = q$
$x_0 = 1$
$q = -2$
$p = 0$
$|p - q + x_0| = |0 + 2 + 1| = 3$

63. B
Sol. $|adj(adj\ A)| = |adj\ A|^2 = (2)^2 = 4$

64. A
Sol. $(z^2 + i)(iz + 1) = 0$
$z = i, \sqrt{-i}, -\sqrt{-i}$
Area = $\frac{1}{2} \times 2 \times \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$



65. C
Sol. $f(g(x)) = x$ and $g(5) = 1$

$$g'(x) = \frac{1}{f'(g(x))} \Rightarrow g'(5) = \frac{1}{f'(1)} = \frac{1}{4}$$

$$g''(x) = \frac{-f''(g(x)) \cdot g'(x)}{(f'(g(x)))^2}$$

$$g''(5) = \frac{-f''(1) \cdot \frac{1}{4}}{(f'(1))^2} = -\frac{3}{32}$$

66. A
Sol. $e^{-\frac{x^2}{2}} > e^{-x^2}$. I_4 is largest

67. D
Sol. $\alpha = 3\cos\theta$, $\gamma = 3\sin\theta$, $\beta = \cos\phi$, $\delta = \sin\phi$
$3(\alpha\beta - \gamma\delta) + 4(\alpha\delta + \beta\gamma) = 3(3\cos\theta\cdot\cos\phi - 3\sin\theta\cdot\sin\phi) + 4(3\cos\theta\cdot\sin\phi + \cos\phi\cdot 3\sin\theta)$
$= 3(3\cos(\theta + \phi) + 4\sin(\theta + \phi))$

68. C
Sol. $\frac{1}{A-1} + \frac{y'}{A+1} = 0$, $A = \frac{y'-1}{y'+1}$

69. B
Sol. $\bar{x} = \frac{1+2+3+4+5}{5} = 3$

$$\sum (x - \bar{x})^2 = 4 + 1 + 0 + 1 + 4 = 10$$

Standard deviation = $\sqrt{\frac{\sum (x - \bar{x})^2}{N}} = \sqrt{\frac{10}{5}} = \sqrt{2}$

70. B

Sol. Volume = $\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix} = 3\alpha\beta\gamma - \alpha^3 - \beta^3 - \gamma^3$

$|v| = (\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 - \alpha\beta - \beta\gamma - \gamma\alpha)$

$= 9(81 - 3 \times 15) = 324$

SECTION – B

71. 2

Sol. $\because\ x^3 + 2x^2 + 2x + 1 = (x + 1)(x^2 + x + 1) = 0$

$x = -1, \omega, \omega^2$

72. 0

Sol. Second curve is parabola with focus (3, –3) and directrix x – y – 2 = 0 equation of axis is x + y = 0 and foot of directrix (1, –1)

Vertex = (2, –2)

$\therefore$ Shortest distance = 0

SECTION – C

73. 00005.00

Sol. Clearly foci are (1, 2) and (5, 5) distance between foci = 5 and transverse axis = 3

$\therefore$ Eccentricity of hyperbola $e = \frac{5}{3}$

$\therefore\ \frac{1}{e^2} + \frac{1}{(e')^2} = 1 \Rightarrow \frac{9}{25} + \frac{1}{(e')^2} = 1 \Rightarrow \frac{1}{(e')^2} = 1 - \frac{9}{25} = \frac{16}{25}$

$e' = \frac{5}{4} \Rightarrow 4e' = 5$

74. 00000.50

Sol. $\lim\limits_{x\to 0}\left(\frac{a^x + b^x}{2}\right)^{\left(\frac{2}{x}\right)} = e^{\lim\limits_{x\to 0}\left(\frac{2}{x}\right)\left(\frac{a^x + b^x}{2} - 1\right)} = e^{\ln a + \ln b} = ab = 6$

(a, b) can be {(1, 6), (6, 1), (2, 3), (3, 2)}

$p = \frac{4}{36} = \frac{1}{9}$; $\frac{9}{2}p = \frac{1}{2} = 0.5$

75. 00001.60

Sol. $2g(x)\cdot g'(x) = g(x)\cdot\frac{2\sec^2 x}{4 + \tan x} \Rightarrow \int g'(x)dx = \int \frac{\sec^2 x}{4 + \tan x}dx$

$\Rightarrow g(x) = \ln(4 + \tan x) + c;\ 0 = \ln 4 + c \Rightarrow c = -\ln 4$

$g(x) = \ln(4 + \tan x) - \ln 4 = \ln\left(1 + \left(\frac{\tan x}{4}\right)\right)$

$g\left(\frac{\pi}{4}\right) = \ln\left(\frac{5}{4}\right)$